

Chapter 1: Neural Networks & Backpropagation

Neural networks are computational models inspired by biological neural systems.

The gradient of the loss function L with respect to weight W is defined as:

$$\frac{\partial L}{\partial W} = \delta \cdot x^T$$

Layer | Neurons | Activation

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Input | 784 | None

Hidden| 128 | ReLU

Output| 10 | Softmax

Chapter 2: Optimization with Gradient Descent

Stochastic gradient descent updates network parameters iteratively:

$$W_{t+1} = W_t - \eta \nabla L(W_t)$$

Learning rate η controls the step size taken towards minimum error.